# Fasting−only Principal Component Analysis

Data scope

- Conditions retained: Baseline (2); NS4 Fasting (18); RTDS Fasting (26).
- NS4 after−meal rows excluded before PCA: 32.
- Fasting rows available: 95; complete PCA rows retained: 67.
- PCA rows excluded (missing/non−positive): 28.

PCA specification

- Variables: 15 measured bile acids only; aggregate classes were not included.
- Preprocessing: LTR median plate adjustment, natural−log transform, centering, and unit−variance scaling.
- Missing values: complete−case matrix across the 15 bile acids; no imputation.
- Algorithm: R prcomp using singular value decomposition.

Interpretation note

PCA is an unsupervised descriptive view of fasting−only multivariate structure.
It complements the fasting mixed−model tests and does not replace those repeated−measures inferences.

PC1 variance: 40.9%
PC2 variance: 21.2%
PC1+PC2 cumulative: 62.0%
PC1+PC2+PC3 cumulative: 74.3%

# What is PCA? — A Beginner's Guide

Principal Component Analysis (PCA) is a statistical technique that simplifies complex, high−dimensional data so that it can be visualised and interpreted more easily.

In this study, every sample has measurements for 15 different bile acids. That means each sample lives in a "15−dimensional space" — one axis for each bile acid. We cannot directly visualise 15 dimensions, so PCA finds new axes (called Principal Components, or PCs) that capture the most important patterns of variation in the data. The first PC (PC1) captures the direction of greatest variation; PC2 captures the next greatest (orthogonal to PC1), and so on.

KEY CONCEPTS

- Score: the position of an individual sample along a principal component axis. Each dot on a PCA score plot is one sample, projected onto the PC1 and PC2 axes.

- Loading: the weight that each original bile acid contributes to a PC. A high loading means that bile acid strongly influences the direction of that PC. Loadings tell you WHICH bile acids are driving the patterns you see in the score plot.

- Variance explained (%): how much of the total variability in the data is captured by each PC. Here PC1 explains 40.9% and PC2 explains 21.2%, so together they capture 62.0% of total variation.

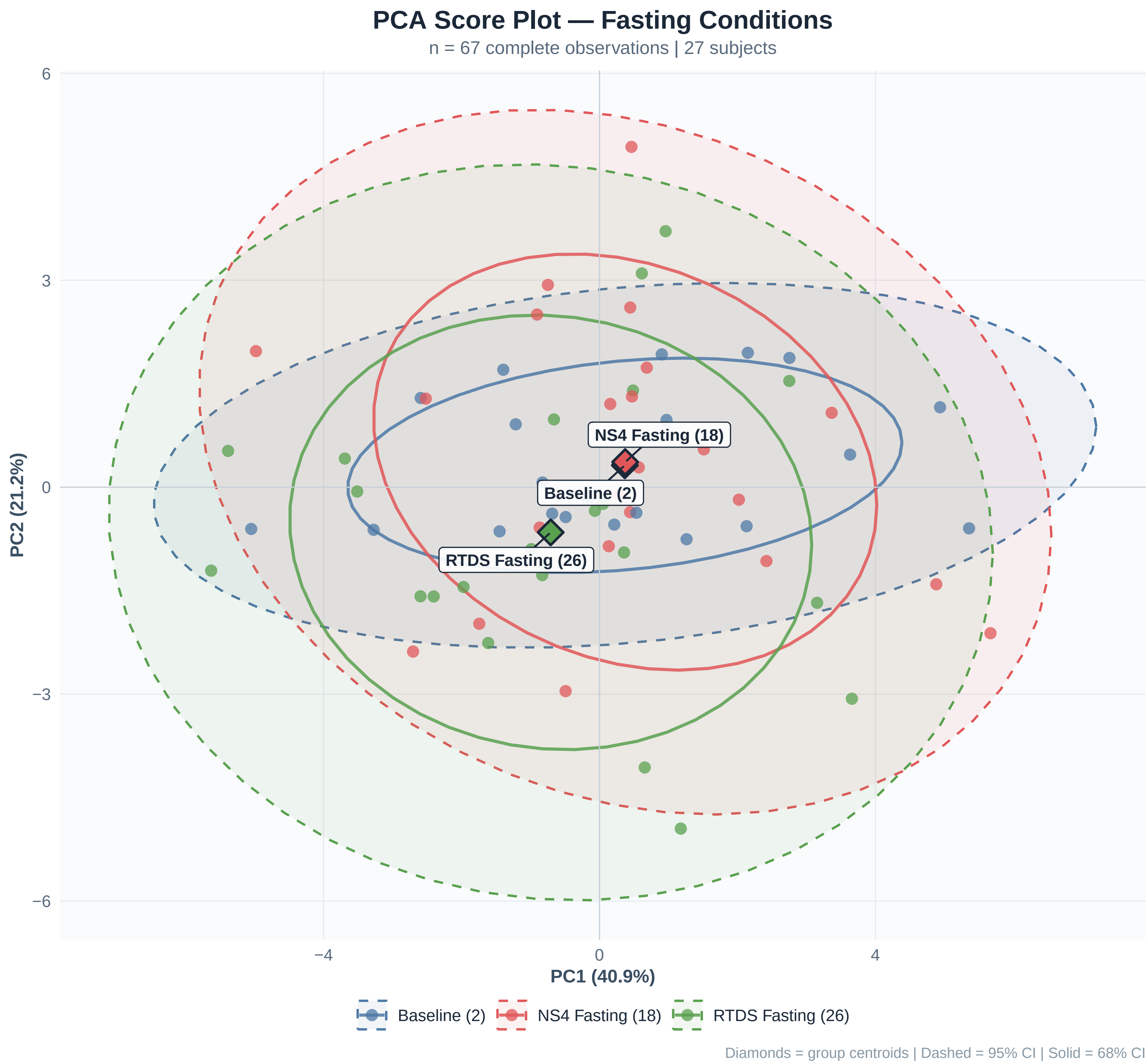
- Centroid: the average position of all samples in a group (e.g. Baseline, NS4 Fasting, RTDS Fasting). If centroids are far apart, the groups have different overall bile acid profiles.

- Ellipse: a confidence region around a group. Overlapping ellipses mean the groups have similar bile acid profiles on those PCs; well−separated ellipses suggest distinct profiles.

- Unsupervised: PCA does not know which samples belong to which group. It only looks at the bile acid values. If groups separate, it means the bile acid profiles themselves naturally differ.

HOW TO READ THE PLOTS THAT FOLLOW

1. Score plot — Look for group separation and overlap. Separation = distinct bile acid profiles.
2. Trajectory plot — Arrows connect the same subject across timepoints, showing individual change.
3. Scree plot — Shows how many PCs are needed; a sharp drop means most info is in the first few PCs.
4. Loading plot — Arrow directions show which bile acids drive each PC direction.
5. Box plots — Are the PC score distributions different between groups?
6. Top loadings / Contributions — Which bile acids matter most for each PC?


PCA Score Plot — Fasting Conditions
n = 67 complete observations | 27 subjects
NS4 Fasting (18)
Baseline (2)
RTDS Fasting (26)
PC2 (21.2%)
PC1 (40.9%)
6
3
0
−3
−6
−4
0
4
Baseline (2)
NS4 Fasting (18)
RTDS Fasting (26)
Diamonds = group centroids | Dashed = 95% CI | Solid = 68% CI

# Interpretation — PCA Score Plot (PC1 vs PC2)

WHAT THIS FIGURE SHOWS

Each dot is one fasting sample positioned according to its overall bile acid profile. Samples that are close together on the plot have similar bile acid profiles; those far apart have different profiles. The x−axis (PC1) captures 40.9% of total variation; the y−axis (PC2) captures 21.2%. Together, these two axes represent 62.0% of all variability across the 15 bile acids.

GROUP CENTROIDS (Diamond markers)

- Baseline (2): PC1 = +0.37, PC2 = +0.32 (n = 21)
- NS4 Fasting (18): PC1 = +0.37, PC2 = +0.36 (n = 23)
- RTDS Fasting (26): PC1 = −0.71, PC2 = −0.65 (n = 23)

HOW TO READ THE ELLIPSES

- Solid ellipses enclose ~68% of each group’s samples (similar to ±1 SD).
- Dashed ellipses enclose ~95% of each group’s samples (similar to ±2 SD).
- When ellipses overlap substantially, the two groups cannot be clearly distinguished on those PCs.
- Non−overlapping ellipses suggest meaningfully different bile acid profiles.

KEY OBSERVATIONS

- Baseline and NS4 Fasting centroids sit at similar PC1 positions, suggesting the overall bile acid “size” (total conjugated BA pool) is comparable between these two conditions.
- The RTDS Fasting centroid shifts to more negative PC1 values, indicating a change in the conjugated bile acid pool under RTDS conditions.
- On PC2, RTDS Fasting moves to more negative values compared with Baseline and NS4 Fasting, pointing to changes in unconjugated/primary bile acids (DCA, CDCA, UDCA, CA — see loading plot).
- The substantial overlap among all three 95% ellipses shows that inter−individual variability is large relative to the condition−level shifts. This is typical for bile acid data.

WHAT THIS MEANS BIOLOGICALLY

PCA did not “know” the condition labels — it only saw bile acid values. The fact that the RTDS centroid is somewhat displaced from Baseline and NS4 suggests a genuine shift in bile acid metabolism under the RTDS diet intervention, even after accounting for the dominant inter−individual variation. However, the heavy ellipse overlap means the effect is modest relative to person−to−person differences, which is why formal mixed−model tests (LMMs) with subject random effects are needed for inference.

Subject Trajectories in PCA Space
Arrows connect repeated measures per subject (n = 23 subjects with =2 timepoints)
5.0
2.5
0.0
−2.5
−5.0
PC2 (21.2%)
−6
−3
0
3
6
PC1 (40.9%)
Baseline (2)
NS4 Fasting (18)
RTDS Fasting (26)
Grey arrows: Baseline −> NS4 Fasting −> RTDS Fasting within each subject

# Interpretation — Subject Trajectory Plot

## WHAT THIS FIGURE SHOWS

This plot uses the same PC1 vs PC2 axes as the score plot, but now thin grey arrows connect the repeated measurements of each subject across timepoints. Of the 27 subjects, 23 had observations at two or more fasting timepoints. Arrows generally flow: Baseline (2) −> NS4 Fasting (18) −> RTDS Fasting (26).

## HOW TO READ IT

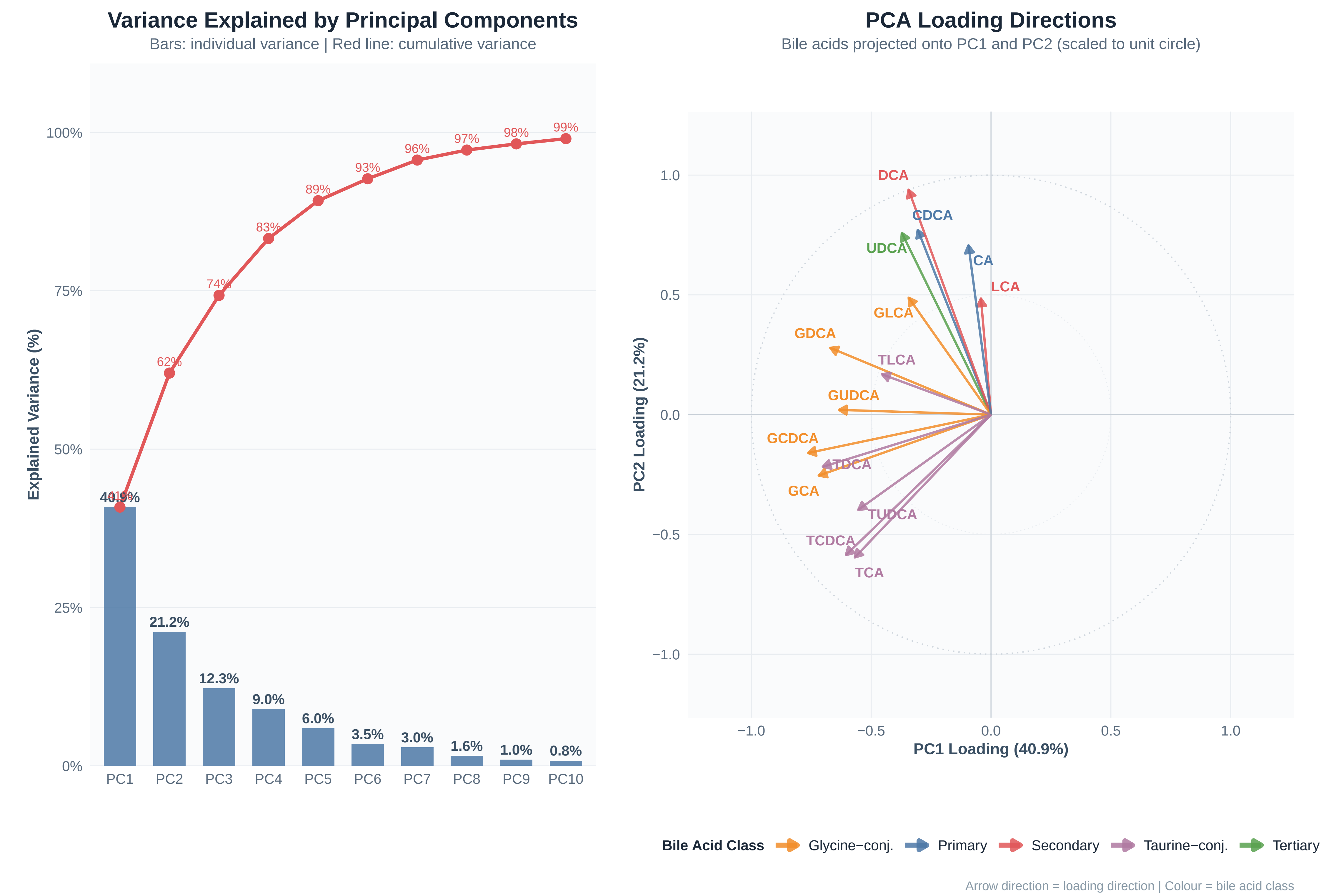
- Each arrow represents one subject’s change in overall bile acid profile between timepoints.
- If all arrows point in the same direction, the dietary conditions cause a consistent shift.
- If arrows point in many different directions, the response to the diet varies greatly between individuals.
- Long arrows = large changes in bile acid profile; short arrows = little change.

## KEY OBSERVATIONS

- Arrow directions are NOT uniform: some subjects move left on PC1 (increasing conjugated BAs), while others move right (decreasing conjugated BAs). Similarly, on PC2 some move up and others down. This heterogeneity reflects high inter−individual variability in bile acid responses.
- Despite this variability, there is a slight net tendency for arrows to drift toward lower (more negative) PC1 and lower PC2 when transitioning to RTDS Fasting, which is consistent with the centroid shift observed in the score plot.
- Some subjects show very large movements (long arrows), suggesting they are strong responders to the dietary interventions, while others barely move.

## WHAT THIS MEANS

The trajectory plot demonstrates why a repeated−measures (mixed−model) analysis is essential: subjects serve as their own controls, and the within−subject changes are what drive statistical power. The individual variability visible here is exactly what the random intercept in the LMM accounts for.
The plot also helps identify potential outlier subjects whose bile acid profiles change dramatically.



Variance Explained by Principal Components
Bars: individual variance | Red line: cumulative variance
Explained Variance (%)
100%
75%
50%
25%
0%
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
40.9%
21.2%
12.3%
9.0%
6.0%
3.5%
3.0%
1.6%
1.0%
0.8%
62%
74%
83%
89%
93%
96%
97%
98%
99%
PCA Loading Directions
Bile acids projected onto PC1 and PC2 (scaled to unit circle)
PC2 Loading (21.2%)
PC1 Loading (40.9%)
1.0
0.5
0.0
−0.5
−1.0
DCA
CDCA
UDCA
CA
LCA
GLCA
GDCA
TLCA
GUDCA
GCDCA
TDCA
GCA
TUDCA
TCDCA
TCA
Bile Acid Class
Glycine−conj.
Primary
Secondary
Taurine−conj.
Tertiary
Arrow direction = loading direction | Colour = bile acid class

# Interpretation — Scree Plot (Variance Explained)

WHAT THIS FIGURE SHOWS

The scree plot (left panel) shows how much of the total variation in the 15 bile acids is captured by each principal component. Blue bars show individual variance per PC; the red line shows the cumulative total.

KEY NUMBERS

- PC1 alone captures 40.9% — this is the single most informative axis.
- PC2 captures 21.2% — the second most important direction.
- PC3 captures 12.3% — still meaningful, but notably less than PC1 or PC2.
- The first 3 PCs together capture 74.3% of all variability.
- The first 5 PCs capture 89.2%, at which point the remaining PCs add very little.

HOW TO INTERPRET THE “ELBOW”

In a scree plot, you look for an “elbow” — a point where the bars drop sharply and then flatten out. Here, there is a clear drop after PC1 (40.9% −> 21.2%) and a further drop after PC3 (12.3% −> 9.0%). This “elbow” at PC3 suggests that the first 3 PCs capture the major patterns and the remaining 12 PCs mostly describe noise or minor individual−level variation.

WHAT THIS MEANS

PC1 alone explains 40.9% of variation, which is quite high for 15 variables.
This means many bile acids are correlated with each other (they rise and fall together), which is biologically expected because bile acids share biosynthetic pathways. The fact that 3 PCs capture ~74% tells us that the 15 bile acids are effectively described by about 3 independent “axes” of variation.

LOADING PLOT (right panel) — see next page for detailed interpretation.

# Interpretation — PCA Loading Plot (Biplot Arrows)

WHAT THIS FIGURE SHOWS

Each arrow represents one of the 15 bile acids. The arrow's direction and length show how strongly and in which direction that bile acid contributes to PC1 (x−axis) and PC2 (y−axis).
Arrows are scaled to a unit circle so that relative directions are easy to compare.

HOW TO READ LOADING ARROWS

- Arrow direction: bile acids pointing in the same direction are positively correlated (they tend to rise and fall together). Arrows pointing in opposite directions are negatively correlated.
- Arrow length: longer arrows are better represented on these two PCs. Short arrows contribute more to other PCs (e.g. PC3 or later).
- Arrows near the outer circle contribute strongly; those near the centre contribute weakly.
- Colour indicates bile acid class: Primary (blue), Secondary (red), Tertiary (green), Glycine−conjugated (orange), and Taurine−conjugated (purple).

KEY OBSERVATIONS

PC1 (40.9% variance) — All 15 arrows point in the same (negative) PC1 direction.
This means PC1 is a "size" axis: when a sample has high overall bile acid levels, it scores high on PC1 (or low, depending on sign convention). The strongest PC1 drivers are the glycine− and taurine−conjugated bile acids: GCDCA, GCA, TDCA, GDCA, GUDCA.

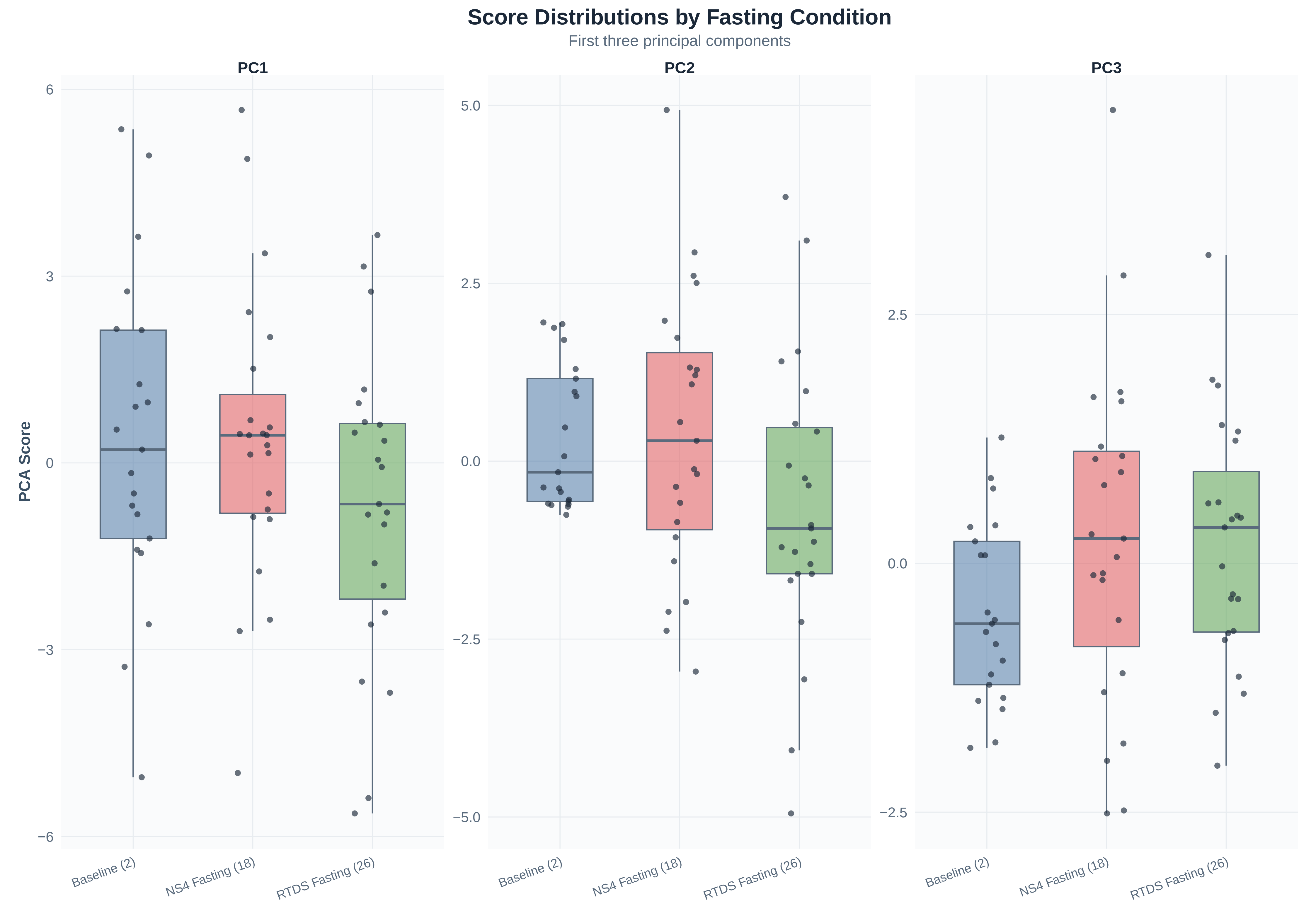
PC2 (21.2% variance) — PC2 separates unconjugated/primary bile acids (DCA, CDCA, UDCA, CA pointing upward) from taurine−conjugated forms (TCDCA, TCA pointing downward). This axis captures the balance between free (unconjugated) and taurine−conjugated species.

LINKING SCORES AND LOADINGS

- A sample that scores high (positive) on PC2 has relatively more DCA, CDCA, UDCA, and CA.
- A sample that scores low (negative) on PC2 has relatively more taurine−conjugated BAs.
- The RTDS Fasting centroid's shift toward lower PC2 suggests a relative increase in taurine−conjugated bile acids (or decrease in free unconjugated species) under RTDS fasting conditions.

CLUSTER OF CONJUGATED SPECIES

Notice how the glycine−conjugated BAs (GCDCA, GUDCA, GDCA, GCA) cluster together and point in a similar direction, as do the taurine−conjugated BAs (TCDCA, TCA, TDCA, TUDCA). This clustering reflects shared metabolic regulation within each conjugation class.


Score Distributions by Fasting Condition
First three principal components
PC1
PC2
PC3
PCA Score
6
3
0
−3
−6
5.0
2.5
0.0
−2.5
−5.0
2.5
0.0
−2.5
Baseline (2)
NS4 Fasting (18)
RTDS Fasting (26)

# Interpretation — Score Distributions by Condition (Box + Strip Plots)

WHAT THIS FIGURE SHOWS

For each of the first three PCs, box plots display the distribution of PCA scores for the three fasting conditions. Each dot is one sample. The box spans the interquartile range (IQR, middle 50%), the line inside the box is the median, and the spread of the dots shows overall variability.

HOW TO READ BOX PLOTS

- If a group's box sits higher or lower than another group's, that group has systematically different scores on that PC — meaning its overall bile acid profile differs in the direction that PC captures.
- Wider boxes (or more scattered dots) indicate greater variability within a condition.
- Overlap between boxes means the effect is modest compared with individual variation.

PC1 DISTRIBUTIONS

- Baseline and NS4 Fasting show similar median PC1 scores, indicating comparable overall bile acid "pool size" between these two conditions.
- RTDS Fasting appears shifted slightly toward more negative PC1 scores, suggesting lower overall conjugated bile acid levels under RTDS conditions.
- All three groups have wide spreads on PC1, confirming that inter−individual variability is the dominant source of variation.

PC2 DISTRIBUTIONS

- RTDS Fasting shows a downward shift compared with Baseline and NS4 Fasting, suggesting a change in the balance between unconjugated (DCA, CDCA, UDCA, CA) and taurine−conjugated species.
- NS4 Fasting and Baseline overlap almost completely on PC2.

PC3 DISTRIBUTIONS

- Baseline appears slightly shifted to negative PC3 scores, while NS4 and RTDS are closer to zero.
- Recall PC3 is driven by GLCA (negative), TLCA (negative), CA and CDCA (positive), so differences on PC3 relate to the balance between lithocholic acid conjugates and primary unconjugated BAs.

OVERALL INTERPRETATION

The box plots reinforce that the largest source of variation in the data is between individuals (wide boxes), not between conditions. The RTDS Fasting condition shows the most visible displacement on PC1 and PC2 relative to Baseline, while NS4 Fasting is largely indistinguishable from Baseline on these PCs. This is consistent with RTDS having a greater impact on the fasting bile acid profile.


Top Bile Acid Loadings by Principal Component
Largest eight signed loadings (absolute value) for PC1–PC3
PC1
PC2
PC3
TDCA
TUDCA
GDCA
TCA
TCDCA
GUDCA
GCA
GCDCA
DCA
CDCA
CA
UDCA
LCA
GLCA
TCA
TCDCA
CDCA
CA
UDCA
LCA
GLCA
TDCA
GDCA
TLCA
−0.50
−0.25
0.00
0.25
0.50
Loading Value
Negative
Positive

# Interpretation — Top Bile Acid Loadings by Principal Component

WHAT THIS FIGURE SHOWS

For each of the first three PCs, horizontal bars display the signed loading of the eight bile acids with the largest absolute loadings. Green bars = positive loading, red bars = negative loading.

HOW TO READ SIGNED LOADINGS

- A positive loading means the bile acid increases in the positive direction of that PC.
- A negative loading means the bile acid increases in the negative direction of that PC.
- Larger bars (in either direction) = more influential bile acid for that PC.

PC1 — OVERALL BILE ACID POOL (“SIZE FACTOR”)

All loadings on PC1 are negative, meaning all 15 bile acids load in the same direction.
Top contributors: GCDCA, GCA, TDCA, GDCA, GUDCA.
These are predominantly glycine− and taurine−conjugated species. PC1 essentially captures the total bile acid pool size: samples with high concentrations across the board score at one extreme, and samples with low concentrations score at the other. This “size” factor is common in metabolomics PCA and typically reflects overall hepatic bile acid production and intestinal recycling.

PC2 — UNCONJUGATED vs. TAURINE−CONJUGATED BALANCE

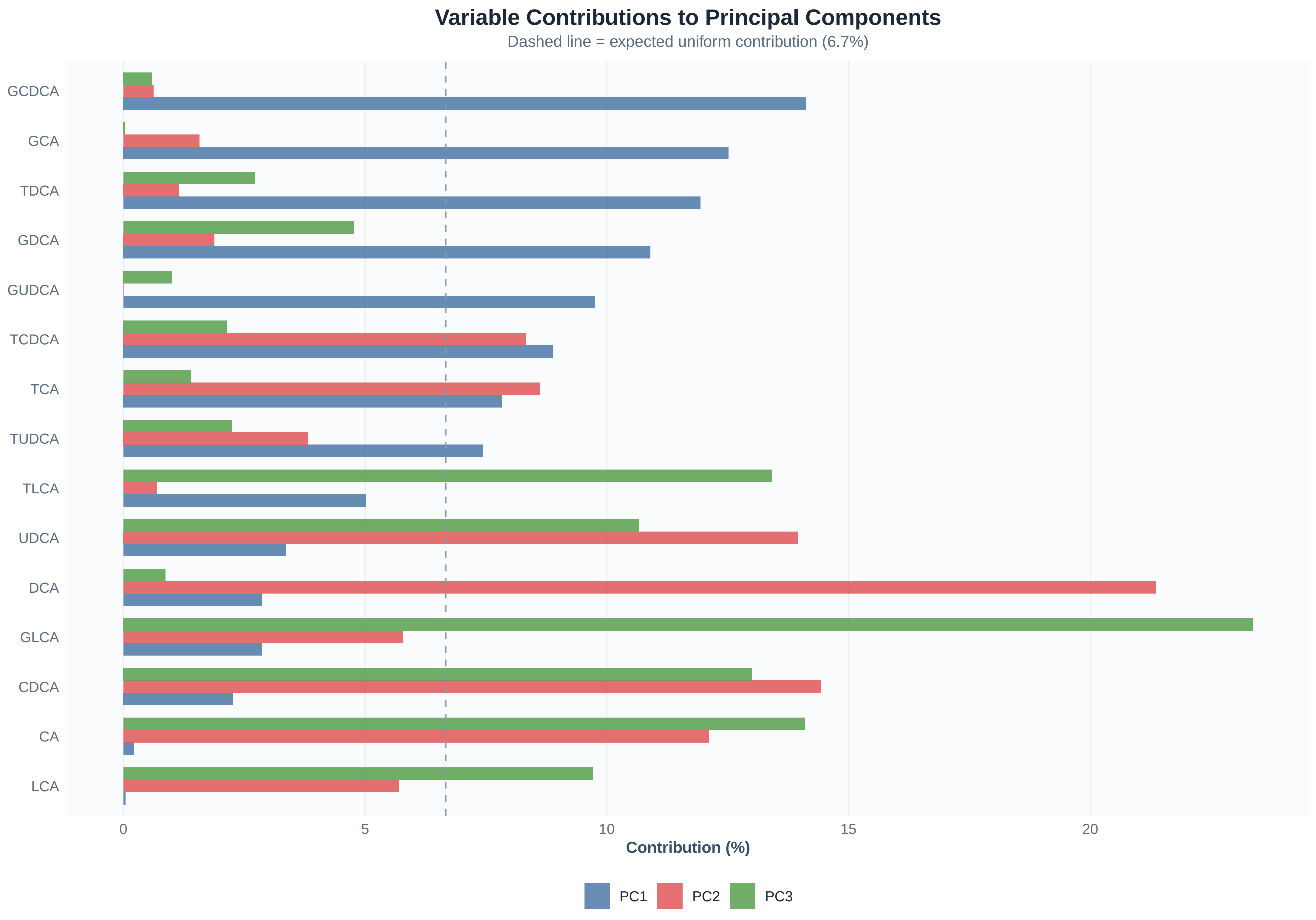
Top positive loaders: DCA, CDCA, UDCA, CA (unconjugated and primary).
Top negative loaders: TCA and related taurine−conjugated species.
PC2 contrasts free (unconjugated) bile acids against their taurine−conjugated counterparts. Samples high on PC2 have a bile acid pool enriched in free species; samples low on PC2 are enriched in taurine conjugates. Shifts on this axis may reflect changes in hepatic conjugation efficiency or gut bacterial deconjugation activity.

PC3 — LITHOCHOLIC CONJUGATES vs. PRIMARY UNCONJUGATED

Top negative loaders: GLCA, CA (glycine− and taurine−lithocholic acid).
Top positive loaders: TLCA, CDCA, UDCA (CA, CDCA, UDCA — primary/tertiary unconjugated).
PC3 captures a secondary contrast between lithocholic acid conjugates (formed by gut bacteria) and the hepatic primary bile acids. This axis may reflect the degree of secondary/tertiary bile acid production by the gut microbiome.

PRACTICAL SIGNIFICANCE

The loading patterns show that PC1 is dominated by conjugated BAs (especially glycine−conjugated), suggesting that the total conjugated bile acid pool is the single most variable feature across samples. PCs 2 and 3 then capture more specific metabolic balances within the bile acid pool.



Variable Contributions to Principal Components
Dashed line = expected uniform contribution (6.7%)
GCDCA
GCA
TDCA
GDCA
GUDCA
TCDCA
TCA
TUDCA
TLCA
UDCA
DCA
GLCA
CDCA
CA
LCA
0
5
10
15
20
Contribution (%)
PC1
PC2
PC3

# Interpretation — Variable Contributions to Principal Components

WHAT THIS FIGURE SHOWS

For each bile acid, grouped bars show its contribution (%) to PC1 (blue), PC2 (red), and PC3 (green). Contribution = squared loading × 100, so it always ranges from 0% to 100% and represents the share of a PC's variance attributable to that bile acid.

The dashed line at 6.7% marks the expected contribution if all 15 bile acids contributed equally to a PC. Bars ABOVE the dashed line contribute more than their "fair share"; bars BELOW contribute less.

HOW TO INTERPRET CONTRIBUTIONS

- A high contribution means the bile acid is an important driver of that PC.
- If one or two bile acids dominate a PC, that PC is essentially capturing variation in those specific bile acids rather than a broad pattern.
- If contributions are evenly spread, the PC reflects a broadly shared pattern across many bile acids.

PC1 CONTRIBUTIONS (blue bars)

- GCDCA, GCA, GDCA, GUDCA, and TDCA are the top contributors (each >10%).
- DCA, CDCA, CA, and LCA contribute very little to PC1.
- This confirms PC1 is driven by conjugated bile acids and represents the conjugated BA pool.

PC2 CONTRIBUTIONS (red bars)

- DCA dominates (~21%), followed by CDCA (~14%), UDCA (~14%), and CA (~12%).
- These are all unconjugated or primary species, confirming PC2 captures unconjugated BA variation.

PC3 CONTRIBUTIONS (green bars)

- GLCA dominates (~23%), followed by CA (~14%), CDCA (~13%), TLCA (~13%), and UDCA (~11%).
- PC3 contrasts lithocholic acid conjugates against primary/tertiary unconjugated species.

PRACTICAL USE

The contribution chart helps decide which bile acids to focus on when discussing condition−level differences. Since PC1 separates samples primarily by their conjugated BA pool, and the RTDS Fasting centroid is displaced on PC1, the conjugated bile acids (GCDCA, GCA, GDCA, GUDCA, TDCA) are the prime candidates for driving the RTDS−associated shift. Cross−reference these with the LMM pairwise results to confirm whether the formal statistical tests agree with the PCA picture.

# Summary — Key PCA Findings and Take−Home Messages

1. DIMENSIONALITY REDUCTION
The 15 bile acids are well summarised by 3 principal components (74.3% variance).
PC1 alone captures 40.9%, indicating strong co−variation among bile acids.

2. PC1 = OVERALL CONJUGATED BILE ACID POOL
All 15 bile acids load in the same direction on PC1, with the strongest contributions from glycine− and taurine−conjugated species (GCDCA, GCA, GDCA, GUDCA, TDCA). PC1 acts as a “size” factor capturing total bile acid abundance.

3. PC2 = UNCONJUGATED vs. TAURINE−CONJUGATED BALANCE
DCA, CDCA, UDCA, and CA load positively; taurine−conjugated BAs load negatively. PC2 reflects the equilibrium between free and taurine−conjugated species.

4. PC3 = LITHOCHOLIC CONJUGATES vs. PRIMARY BAs
GLCA and TLCA (secondary, microbiome−derived) contrast with CA, CDCA, and UDCA (primary/tertiary).

5. GROUP SEPARATION IS MODEST BUT PRESENT
- Baseline and NS4 Fasting overlap substantially: their bile acid profiles are similar.
- RTDS Fasting shows a visible centroid shift (more negative on PC1 and PC2), suggesting a change in the overall conjugated BA pool and the unconjugated/taurine−conjugated balance.
- However, inter−individual variation dominates: subject−to−subject differences are much larger than condition−level differences.

6. INDIVIDUAL RESPONSE HETEROGENEITY
The trajectory plot shows subjects respond to the dietary interventions in different directions and magnitudes, reinforcing the importance of within−subject repeated−measures modelling (LMMs).

7. RELATIONSHIP TO FORMAL STATISTICAL TESTS
PCA is unsupervised and exploratory. It DOES NOT test hypotheses or compute p−values. The patterns observed here should be validated against the fasting−only linear mixed model (LMM) results, where subject−level random effects properly account for repeated measures and formal tests control for multiplicity via FDR correction.

Specifically: if the LMM detects significant pairwise differences for conjugated bile acids (e.g. GCDCA, GDCA, TDCA) between RTDS Fasting and Baseline, this would be consistent with the PCA finding that RTDS shifts the conjugated BA pool (PC1). Similarly, significant changes in unconjugated BAs (DCA, CDCA, UDCA) would confirm the PC2 pattern.

# PCA Output Files

PDF report: fasting_only_analysis/outputs/Fasting_Only_PCA_Report.pdf

CSV outputs (in
fasting_only_analysis/outputs/pca
/)

- fasting_pca_scores.csv — PC scores for each retained observation
- fasting_pca_loadings.csv — Variable loadings for all PCs
- fasting_pca_variance.csv — Variance explained per PC
- fasting_pca_contributions.csv — Variable contributions (squared loadings)
- fasting_pca_input_log_complete_cases.csv — Log−transformed input matrix
- fasting_pca_missingness.csv — Missing/non−positive row counts per compound
- fasting_pca_verification_checks.csv — Data audit trail

PNG plots (in
fasting_only_analysis/outputs/pca/plots
/)

- pca_scores_plot.png — Score plot with ellipses and centroids
- pca_subject_trajectories.png — Paired−subject movement in PCA space
- pca_scree_plot.png — Variance explained summary
- pca_loadings_plot.png — Loading directions with bile acid classes
- pca_score_distributions.png — Box plots of PC scores by condition
- pca_top_loadings.png — Top signed loadings for PC1–PC3
- pca_variable_contributions.png — Variable contribution percentages